

Soham Mukherjee

Personal information

Date of Birth *February 6th, 1998*
Place of Birth *Kolkata, India*
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Education

2022–2026 **Ph.D, Faculty of Science**, University of Tübingen, Tübingen, Germany (International Max Planck Research School for Intelligent Systems).
2019–2021 **Master of Science in Statistics**, *School of Mathematics and Statistics*, University of Hyderabad, Hyderabad, India.
CGPA (out of 10): 8.56 Grade: First Division with Distinction
2016–2019 **Bachelor of Science (Honours) in Statistics**, *Bidhannagar College, WBSU*, Kolkata, India.
Percentage: 69.88% Grade: First Class Honours

Work experience

Mar 2026–Current **Postdoctoral Researcher**, *Department of Computer Science, Aalto University*, Espoo, Finland.
o Bayesian workflow, model comparison, assessment and diagnostics
Jan 2026–Mar 2026 **Academic Scientist**, *M3 Forschungszentrum, Universitätsklinikum Tübingen*, University of Tübingen, Tübingen, Germany.
o On latent variable estimation using derivative Gaussian processes (*Ph.D Thesis*)
Jan 2024–Dec 2025 **Academic Scientist**, *Faculty of Statistics, TU Dortmund*, Dortmund, Germany.
o On latent variable estimation using derivative Gaussian processes (*Ph.D Thesis*)
Aug 2022–Dec 2023 **Academic Scientist**, *Innere Medizin I, Universitätsklinikum Tübingen*, University of Tübingen, Tübingen, Germany.
o On latent variable estimation using derivative Gaussian processes (*Ph.D Thesis*)
Feb 2021– Oct 2021 **Research Intern**, *Indian Statistical Institute*, Kolkata, India.
o On Bayesian stochastic volatility model using finite Gaussian mixtures with applications to high-frequency NSE-Nifty stock returns.
Jul 2020– Aug 2020 **Summer Research Trainee**, *Council Of Scientific And Industrial Research–North East Institute Of Science And Technology*, Jorhat, Assam, India.
o Regression analysis and prediction algorithm for various atmospheric target variables for cloud formation inference.

- May 2020– Jun 2020 **Summer Intern, Deloitte India (Offices of US)**, Hyderabad, India.
- o Project based on text analysis, text pre-processing and sentiment analysis using Natural Language Processing algorithms.
 - o Project based on exploratory data analysis and predictive models using Linear Models, CART and Random Forest.

Supervision

- 2023–2024 Marcello Zago
- o **M.Sc. thesis:** Derivative Gaussian processes for Single-Cell RNA sequencing data using combined covariance functions.

Teaching

- 2025 Seminar on model comparison: Summer semester (TU Dortmund)
- 2024 Seminar on multilevel models: Winter semester (TU Dortmund)
- 2024 Intensive course in statistics: Summer semester (TU Dortmund)

Conference/workshops

- 2025 **Royal Statistical Society International Conference 2025 (RSS 2025):** contributed talk on "DGP-LVM: Derivative Gaussian process latent variable models."
- 2024 **International Conference on Statistics and Data Science (ICSIDS 2024):** contributed talk on "DGP-LVM: Derivative Gaussian process latent variable models."
- 2024 **International Society for Bayesian Analysis:** attended
- 2023 **Cambridge Ellis Unit Summer School on Probabilistic Machine Learning 2023:** poster presentation on "Derivative Gaussian processes latent variable models with applications to Single-Cell RNA sequencing data."
- 2021 Lectures in Statistics, Indo-French Workshop organized by IFCAM and ISI
- 2020 Prof. C.R. Rao Birth Centenary Conference on Statistics and Applications, 2020 funded by DST-SERB, Govt. of India.
- 2017 MTUSS-2017 (Mathematics Training for Undergraduate Statistics Students) programme comprising courses in Foundations, Real Analysis, Linear Algebra and Probability funded by NBHM.

Theses and other projects

- Dec 2025 Ph.D. Thesis
- o On latent variable estimation using derivative Gaussian processes.
- July 2021 M.Sc Thesis
- o Research and modelling of Supermassive black hole mass using Bayesian generalized linear mixed models and Bayesian missing value methods.
- May 2020 Worldwide COVID-19 data modelling and forecasting
- o Team project for visualization and forecast of the time series data with ARIMA, Holt-Winters Exponential Smoothing, Polynomial Regression and Support Vector Machines on the effect of the pandemic. <https://www.kaggle.com/soham6298/covid-19-uoh-d5380e>

Feb 2019 B.Sc. Thesis

- o Study of forecasting methods for Stock Market price index: A comparative analysis of ARIMA and Exponential Smoothing methods for stock market price data.

Technical and Language Skills

Software R, Stan, JAGS, Pyro-PPL

Bengali Native proficiency

English TOEFL iBT (Sept 2021): 114/120; Reading: 30, Listening: 28, Speaking: 29, Writing: 27

Hindi Professional proficiency

German 3 years course funded by IMPRS-IS (upto B1)

Research articles

Mukherjee S., Claassen M., & Bürkner P. C. (2026). Hilbert space methods for approximating multi-output latent variable Gaussian processes. *Stat Comput* 36, 110. <https://doi.org/10.1007/s11222-026-10869-x> [Pre-print:<https://arxiv.org/abs/2505.16919>]

Mukherjee S., Aguilar J. E., Zago M., Claassen M. & Bürkner P. C. (2025). Latent variable estimation with composite Hilbert space Gaussian processes.(*in review*)(<https://arxiv.org/abs/2510.25371>)

Bischoff S., Poličar P.G., **Mukherjee S.**, Macke J.H., Claassen M., & Schröder C. (2025). velotest: Statistical assessment of RNA velocity embeddings reveals quality differences for reliable trajectory visualizations.(*in review*)(<https://doi.org/10.1101/2025.10.26.683064>)

Mukherjee S., Claassen M., & Bürkner P. C. (2025). DGP-LVM: Derivative Gaussian process latent variable models. *Stat Comput* 35, 120. <https://doi.org/10.1007/s11222-025-10644-4> [Pre-print: <https://arxiv.org/abs/2404.04074>]

Mukherjee S. (2021) Bayesian Analysis of Stochastic Volatility Model using Finite Gaussian Mixtures with Unknown Number of Components (<https://arxiv.org/abs/2110.12824>)